GEOG 761 Lab 2 | Benjamin McLean

Answer Proforma

(1) What other mathematical compositing approaches could you take when building your single image from the filtered image collection you create in the cell above? Hint: you currently are doing a 'median-composite'. (3 pts)

For compositing from an image collection there are several other mathematical reducers available:

- .mean() -- Average pixel value
- .Min/Max() -- returns min or max pixel values for each band, useful for indexing
- .Standard Deviation() -- useful for change detection.
- .mode() -- most frequently occurring pixel value.
- .sum() -- sum of pixel values, used for accumulated totals
- ee.Reducer.percenile() -- for a specified percentile giving more control over the value distribution.
- .qualityMosaic() -- for selecting pixels with the best value in a QA band.

(2) What year of data was this landcover map (accessed in the cell above) created on? (1 pt)

Dataset availability via GEE catalogue: 2020-01-01–2021-01-01 (2020)

(3) What type of landcover is being sampled in sample number 2 of the dataset? Make sure your random seed is set to 2 in the sampling code so that you get the same answer as me. (1 pt)

Sample 2 is identified as Class 8 (80) which corresponds to 'Permanent Water Bodies'.

(4) Look at the ESA landcover map and our resulting random forest classified Sentinel 2 image map. How are they different and what might be some causes of this difference? (5 pts)

They use different colours for the classes, with the S2 landcover using a random visualiser. The pixel sizes are the same as they are both derived from S2 imagery, though the S2 classification appears considerably more grainy in certain areas which is likely due to the pixel reflectances being assessed using only a handful of S2 bands (B2,3,4,8), resulting in reduced accuracy compared to the ESA product which used both Sentinel-1 radar data and Sentinel-2 data in addition to spectral indicies like NDVI and whatever else they used according to the product manual. Generally though, the S2 classification has captured fairly similar areas for key classes such as urban, grassland, forest, and water.

(5) Exercise: modify the code so that you feed into your random forest classifier that has been trained on 2020 data, on other years of Sentinel 2 data than 2020. Produce a publication quality figure that presents the following:

- Landcover maps of Wellington for 2018, 2022 and 2024
- The Test accuracy averages of the RF classifier for 2020.
- Include a sentence in your figure caption that explains why you cannot state the accuracy of the classifier for years other than 2020.

(15 pts)

Landcover Classification of Wellington, New Zealand (2018-2024)

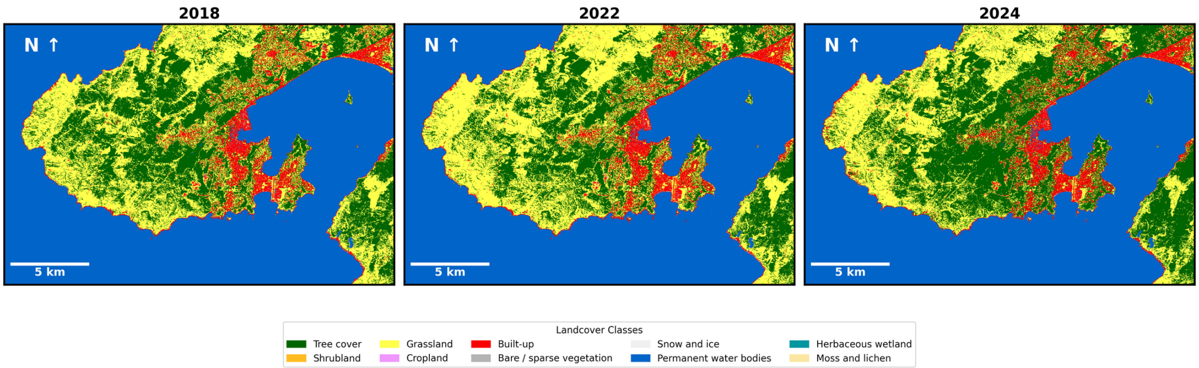

Figure 1: Multi-temporal landcover classification maps of Wellington for 2018, 2022, and 2024, derived using a Random Forest classifier trained on 2020 Sentinel-2 data. For the 2020 validation, the classifier achieved an Overall Accuracy of 0.90 and a Weighted Avg F1-score of 0.90. We cannot state the accuracy for years other than 2020 because the ground truth reference data (ESA WorldCover) was only available for the year of 2020.

(6) Exercise: using an SVM model and the Landcare NZ 2024 landcover database, produce a landcover map of Great Barrier (Aotea) Island for 2025 (based off the Austral summer of 24/25).

Your map should be presented at a publication quality level with all the usual map components (scale, legend, north arrow, data attribution).

You will need to provide performance statistics of the model within your figure.

- Here you can access the landcover database: https://lris.scinfo.org.nz/layer/104400-lcdb-v50-land-cover-database-version-50-mainland-new-zealand/. You will need to explore for yourself how to extract this data and then upload it to colab, then how to plug it into the SVM algorithim. I have provided some starter code below.

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An intial workflow to get the data into the state you need it in to then use it as training data might look like:

- Download the ZIP manually from their browser, having set your area of interest and used the 'Export' tool top right.
- Upload it to Colab.
- Unzip it and load with GeoPandas. (25 pts)

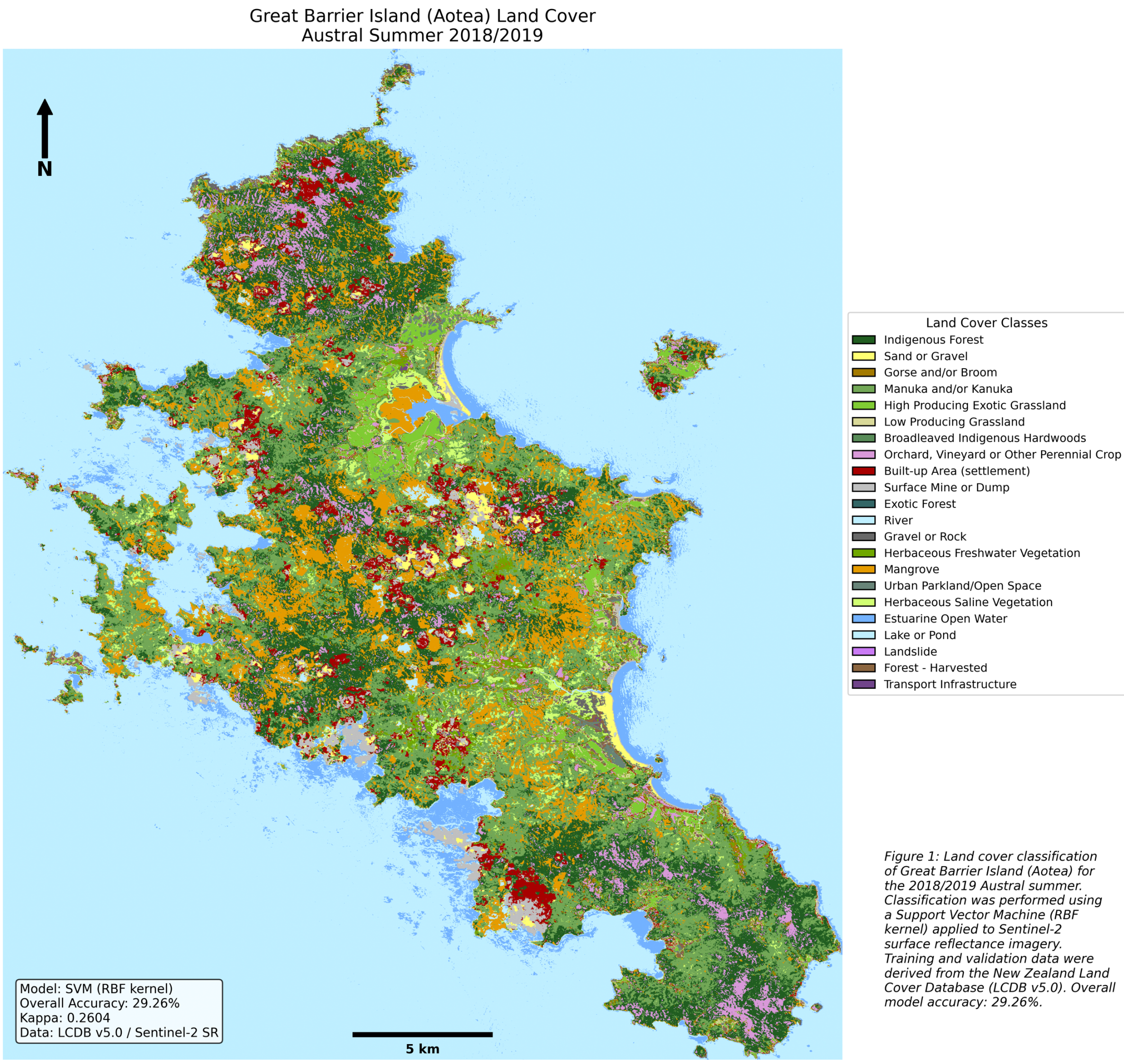

Figure 1: Land cover classification of Great Barrier Island (Aotea) for the 2018/2019 Austral summer. Classification was performed using a Support Vector Machine (RBF kernel) applied to Sentinel-2 surface reflectance imagery. Training and validation data were derived from the New Zealand Land Cover Database (LCDB v5.0). Overall model accuracy: 29.26%.